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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 12: Assignment 12 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) You have been recruited as a lead engineer by ArrEll corporation which wants to enter the self-driving car market. In the context of the standard Reinforcement Learning framework, what would you classify as the **state** and **actions**? Note that your system does not have access to previous states. **1 point**

- ☐ **State**-(Current steering wheel position, Current pedal positions, Current speed) **Actions**-(Turn the wheel, Press pedals)
- ☐ **State**-(Current steering wheel position, Current pedal positions, Current acceleration) **Actions**-(Turn the wheel, Press pedals)
- ☐ **State**-(Current steering wheel position, Current pedal positions, Current speed) **Actions**-(Change direction, Change speed)
- ☐ **State**-(Current steering wheel position, Current pedal positions, Current acceleration) **Actions**-(Change direction, Change speed)

2) Which of these classifiers do not require any additional modifications to use them when we have more than 2 classes? (Note that more than one answers could be correct) **1 point**

- ☐ Decision Trees
- ☐ Linear Classifier
- ☐ Support Vector Machines (SVMs)
- ☐ k Nearest Neighbors (kNN)

3) You are designing a Reinforcement Learning agent for a racing game. Among the following reward schemes, which one leads to the best performance of the agent? **1 point**

Week 10 ()**Week 11 ()****Week 12 ()**

- ☐ Learning Theory (unit? unit=129&lesson=130)
- ☐ Introduction to Reinforcement Learning (unit? unit=129&lesson=131)
- ☐ RL Framework and TD Learning (Optional) (unit? unit=129&lesson=132)
- ☒ Solution Methods and Applications (Optional) (unit? unit=129&lesson=133)
- ☐ **Practice: Week 12: Assignment 12 (Non Graded) (assessment? name=275)**
- ☐ Week 12 Feedback Form : Introduction To Machine Learning (unit? unit=129&lesson=293)
- ☒ Quiz: Week 12 : Assignment 12 (assessment? name=304)
- ☒ Week 12: Solution (unit?

- ☐ +5 for reaching the finish line, -1 for going off the road
- ☐ +5 for reaching the finish line, -0.1 for every second that passes before the agent reaches the finish line
- ☐ +5 for reaching the finish line, -0.1 for every second that passes before the agent reaches the finish line, +1 for the agent going off the road.
- ☐ -5 for reaching the finish line, +0.1 for every second that passes before the agent reaches the finish line.

Check Answers and Submit



unit=129&less
on=305)

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July 2024 ()**

